Problem: A Platform Related to Identify AI-Generated or AI-Altered Videos/Images and Trace Their Origin Across Social Media.

Why do we need this technology?

- **Difficulty in Identifying AI Content**
  It is becoming difficult to distinguish between real images/videos and AI-generated content.
- **Deepfake and Manipulated Media**
  AI can realistically manipulate a person's face, voice, image, or video.
- **Spread of Fake Information**
  Fake AI-generated content can rapidly go viral on social media and spread misinformation.
- **Unknown Original Source**
  It is difficult to identify the original source of a viral image or video.
- **No Tracking of Content Spread**
  It is not easy to track where an image/video has been shared across multiple platforms like Facebook, Instagram, X, and YouTube.
- **Loss of Trust**
  It becomes difficult for users to decide whether online content is real, authentic, or trustworthy.
- **Lack of a Unified Solution**
  There is a lack of a simple platform that provides **AI-generated media detection + altered content identification + origin tracing + social media spread tracking** all in one place.

Solution: According to the brochure, the solution should:

- Detect AI-generated images/videos
- Detect AI-altered/manipulated images/videos
- Verify the authenticity of the content
- Trace the origin of that content
- Track its dissemination on social media
- Be used to combat misinformation, impersonation, cyber fraud, and digital evidence tampering.

Core Detection Capabilities

1. Detecting AI-generated images/videos

Meaning: Identifying whether an image or video has been entirely generated by AI or not.

Example: A celebrity video goes viral in which they are giving a controversial statement.

The platform will analyze the video and check for:

- Signs of face generation
- Whether lip-sync is unnatural
- Facial expressions
- Lighting/shadows
- Skin/face artifacts
- Frame-to-frame inconsistencies
- Audio-video synchronization
- Known AI-generation patterns
- Metadata, encoding, etc.
- Then the system may provide a result:

> **AI Generated Probability: 94%**

Important point: 94% does not mean absolute proof. This is the detection model's confidence/probability.

1. Detecting AI-altered / manipulated images/videos

This is slightly different from AI-generated content. Here, the original content may be real, but it has been manipulated.

Example: There is a real photograph: Person A is standing at an event. Someone edits it to:

- Add an object to Person A's hand.
- Or replace the face.
- Or replace one person's face with another person's face in a video.
- Or remove the original audio and add fake audio.

Broadly, this can be understood as manipulated/altered media.

The platform will need to identify:

- Face swap
- Deepfake
- Object insertion/removal
- Background manipulation
- Audio manipulation
- Video frame manipulation
- Image splicing
- AI enhancement/generative editing
- Transformations such as cropping/re-encoding

Example result:

> **Originality:** Likely manipulated
> **Detected modification:** Face replacement
> **Confidence:** 91%

1. Verifying content authenticity

This is a broader concept than just AI detection.

The question is not: "Was AI used?"

The question is: "Does this content genuinely represent what it claims to represent?"

Example: A photo is circulating with the caption: "Today in Chandigarh, police opened fire on protestors."

The image may be real. It may also not be AI-generated. But it could be a photo from 5 years ago, from a different country.

So the AI detector would say: **Not AI-generated.**

But authenticity verification would say: **Claim/context potentially false.**

That is why your platform should combine multiple signals:

Media analysis + metadata + source information + timestamps + contextual verification + provenance.

Origin Tracing, Dissemination & Combat Applications

01

Trace origin

Find the earliest identifiable source of suspicious content across platforms.

02

Track spread

Map how the same content moves from one network to another over time.

03

Combat abuse

Use origin and spread signals to detect misinformation, impersonation, fraud, and tampering.

1. Tracing the origin of content

This is a very important part of Problem 4. When a suspicious video, image, or audio is found on a platform, the first question is: **where did this content first come from?**

The platform's goal is to find the **earliest identifiable instance** based on available public evidence. This does not mean proving the "first-ever source," because deleted posts, private groups, closed accounts, and encrypted channels are usually not visible.

Practical steps

1. Extract matching frames, audio, captions, and metadata from the content.
2. Search for the same or similar content across multiple platforms.
3. Compare upload time, account age, repost chains, and engagement history.
4. Mark the earliest publicly visible instance as the "origin candidate."

What to look for

- First upload time
- Reposts and mirrors
- Watermarks or platform logos
- Metadata, encoding, and file fingerprints
- Account credibility and source history

Example: You receive a viral video. Search results show:

- Video A → uploaded Aug 1
- Video B → uploaded Aug 3
- Video C → uploaded Aug 5

The platform can report: **Earliest discovered source: Video A — Aug 1**

Important: the earliest discovered source and the actual first-ever source are not the same.

2. Tracing dissemination on social media

Origin and **dissemination** are not the same thing. Origin tells us where the content was first found in an identifiable form. Dissemination tells us **how it spread** afterward.

1 **Original post**

The content first appears on one source.

2 **Cross-posting**

The same content is reposted on X, Instagram, Telegram, or Facebook.

3 **Amplification**

Influential accounts, groups, or pages spread the content even faster.

4 **News or wider reach**

The content reaches mainstream coverage or a broader public audience.

The platform can build a **propagation graph** where:

- **Node =** post, account, channel, or website
- **Edge =** repost/share of the same or similar content
- **Direction =** from which source to which destination the spread occurred

Example spread path:

Source → X → Instagram → Facebook → Telegram → News website

This graph allows investigators to see in which direction the content spread, which accounts boosted it, and at what stage viral momentum built up.

Spread analysis signals

- Repost frequency
- Time gaps between shares
- Follower count of amplifying accounts
- Hashtag and caption similarity
- Language or region-based spread

Why it matters

Dissemination analysis shows whether the content was organic or coordinated. If multiple similar reposts appear at the same time, suspicion of a coordinated campaign can become stronger.

3. Combat applications: fighting misinformation

The most practical use of origin tracing and dissemination tracking is in misinformation response. When the platform has source, spread pattern, and context signals together, it has stronger evidence for investigation.

Misinformation detection

Identify false narratives by combining virality, suspicious source history, and content mismatch.

Impersonation detection

Understand whether a fake identity of a public figure, official, or organization is being used.

Cyber fraud prevention

Detect payment, extortion, or scam attempts using fake audio/video/image.

Evidence tampering checks

Use forensic signs to see whether a media file has been edited, re-encoded, or modified.

Practical example: Suppose a fake video goes viral: "The government has implemented a new law today." The platform analyzes:

- AI detection → 96% likely AI-generated
- Source analysis → earliest identifiable source is a suspicious account
- Context analysis → the claimed event date does not match the video metadata
- Dissemination analysis → spreading rapidly across 3 platforms

Now the investigator can receive a consolidated alert: **High-risk manipulated content**

Impersonation example

An AI-generated video of a police officer is created: "There is a warrant for your arrest, make a payment at this number."

The platform can detect face manipulation, voice cloning, lip-sync anomalies, and similar source patterns and raise an alert: **Possible AI impersonation detected.**

Cyber fraud example

A fake video of a company CEO is created: "Transfer ₹20 lakh to the company account." Or ransom is demanded using a fake image/video of a person.

Fake media + identity impersonation + suspicious source + spread pattern can become useful evidence in a cybercrime investigation.

4. Combating digital evidence tampering

This is very important for police and forensic teams. If CCTV or bodycam footage is found during an investigation and frames have been removed, the timestamp has been altered, or the audio has been modified, the platform can raise suspicion of tampering using forensic indicators.

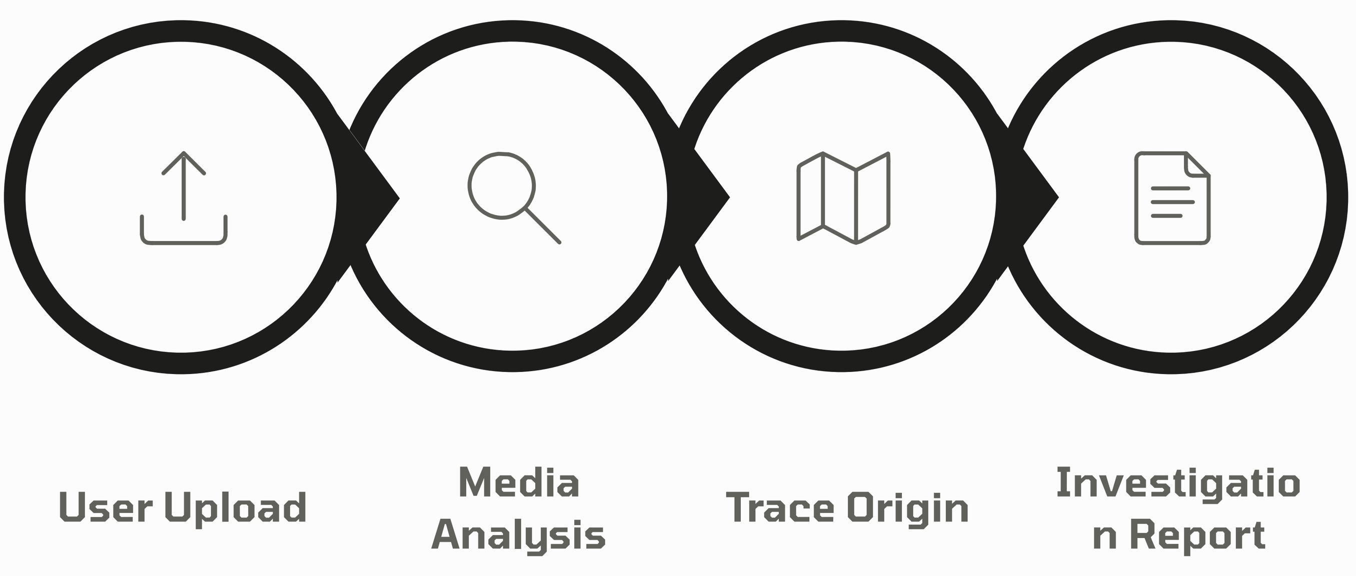
Example: Original-looking video → frame analysis → timestamp inconsistency → compression inconsistency → possible editing detected

⚠ Potential evidence tampering detected

However, in a court context, the system should not make an absolute claim like "100% fake." A better output would be: **evidence + confidence + forensic indicators**.

Conceptualize Format & Complete Development Roadmap

Conceptualize format:



USER UPLOADS IMAGE / VIDEO ↓ Media Analysis ↓ AI Generated Detection AI Manipulated Detection Metadata/Forensics

- AI Generated: 87%
- Manipulated: 92%
- Authenticity Risk: HIGH
- Possible Face Swap: Detected
- Earliest Identifiable Source: Source X
- First Seen: 12 Aug 2026
- Platforms Found: X, Instagram, Facebook, Telegram
- Similarity: 97%
- Dissemination: High
- Risk Category: Impersonation / Misinformation

And at the very bottom:

Evidence Timeline

12 Aug Original identifiable post ↓ 13 Aug X repost ↓ 14 Aug Instagram upload ↓ 15 Aug Facebook repost ↓ 16 Aug Telegram circulation

This is what can turn a simple "Deepfake Detector" into a Digital Investigation & Provenance Platform.

Now the main solution part = Complete Development Roadmap

Upload Image/Video → AI Detection → Forensic Analysis → Origin Tracing → Social Media Spread Analysis → Investigation Report

STEP 1 — Break the problem into modules

First, the project should be divided into 5 major modules:

1. Image Detection | **2. Video Detection** | **3. Forensic Analysis**

4. Origin & Social Media Tracing | **5. Investigator Dashboard + Report**

STEP 2 — Collect the dataset 📊

You need to build a dataset for your AI model:

Real Images + AI-Generated Images + AI-Altered/Manipulated Images

And for video:

Real Videos + AI-Generated/Deepfake Videos + Manipulated Videos

Then divide the dataset into:

Train / Validation / Test

.

STEP 3 — Image Detection Model 🧠

First solve images only.

Flow:

Image ↓ Preprocessing ↓ AI Detection Model ↓ Prediction ↓ Confidence Score

Example:

AI Generated: 93% Real: 7%

Here, the model has to be trained, validated, and tested.

STEP 4 — Image Manipulation Detection

It is not enough to detect only whether an image is AI-generated.

Now the model/pipeline has to identify:

- AI-generated
- Face manipulation
- Object manipulation
- Background manipulation
- Image splicing/editing
- Other suspicious alterations

And, if possible, show the manipulated region/localization as well.

STEP 5 — Video Detection 🎥

Now do not apply the image model directly to video.

The video has to be processed like this:

Video ↓ Frame Extraction ↓ Important Frames ↓ Image Detection ↓ Temporal Analysis ↓ Final Video Result

Example:

Video duration: 20 sec 600 frames analyzed 184 suspicious frames detected Overall probability: 91%

STEP 6 — Video-specific Forensics

Additional signals need to be analyzed in the video:

- Face consistency
- Frame-to-frame artifacts
- Lip/face synchronization
- Temporal inconsistencies
- Audio/video inconsistencies
- Compression anomalies

Here, your system will no longer be just a frame classifier.

STEP 7 — Metadata / Evidence Analysis 🔎

Extract the forensic information from the uploaded evidence:

- Filename
- File type
- File size
- Creation metadata
- Modification metadata
- EXIF
- Encoding information
- Frame information
- Hash

And importantly:

Hash generation

Generate a cryptographic hash of the evidence.

Example:

SHA-256 ↓

a84f....91c2

This can demonstrate evidence integrity during the investigation.

STEP 8 — Origin Tracing 🌐

Now the second major part of the problem statement.

The system has to investigate: Where did this content originate on the internet?

Possible workflow:

Uploaded Evidence ↓ Content Fingerprint ↓ Search / Matching ↓ Potential Sources ↓ Earliest Available Source

It is important here to understand that origin ≠ automatically the true original source.

The system should present evidence in the form of a potential/earliest discovered source.

STEP 9 — Social Media Dissemination Analysis

Now see how the content spread:

Source ↓ Account A ↓ Account B ↓ Account C ↓ Account D

This can be visualized in graph form:

Node = account/post/source

Edge = sharing/reposting relationship

This will help the investigator understand the propagation pattern.

STEP 10 — Investigation Dashboard 🖥️

Now connect all the modules into one website.

Dashboard:

FORENSIC ANALYSIS

Upload Evidence

[IMAGE / VIDEO]

ANALYZE

Result page:

AI STATUS

AI Analysis Results, Origin Dashboard & Final Report

AI Generated Confidence	**93.4%**	**Manipulation** Manipulation Detected

93.4%

AI Detection

Generated confidence

87%

Manipulation

Overall likelihood

1

Primary Source

Earliest discovered origin

5

Platforms

Spread observed across channels

AI Detection Results

Confidence score: 93.4%

Verdict: High likelihood of synthetic or AI-assisted content

Progress indicator: ██████████████████████████░ 93.4%

Manipulation Detection

Type: Face manipulation / metadata anomaly / synthetic artifacts

Confidence: High

- Synthetic artifacts detected
- Metadata anomaly detected
- Face consistency issues flagged

Origin Information

Source: Source A

First appearance: Earliest discovered post

Evidence trail: Earliest discovered source → Post B → Post C → Post D

Graph, timestamps, and platform URLs can be attached where available.

Dissemination Map

01
Source A

02
Post B

03
Post C

04
Post D

05
Platforms

Track how the content spread across platforms, accounts, and repost chains to visualize dissemination patterns.

Automated Forensic Report 📄

- CASE ID
- EVIDENCE HASH
- FILE INFORMATION
- Classification
- Confidence
- Model version

Final Verdict

⊗ High-risk content: likely synthetic or manipulated and should be reviewed by an investigator before use.

The full project in one line

01
Dataset

02
Image Model

03
Image Forensics

04
Video Pipeline

05
Metadata + Evidence Integrity

06
Origin Tracing

07
Social Media Dissemination

08
Backend API

09
Web Dashboard

10
Automated Report

11
Testing + Evaluation

12
Hackathon Demo

Our Team

Present by - Pratham

video link=
https://www.loom.com/share/cc96be14eb274ba6b82beacf2dcfa959